Orientation Tracking for a Rigid Body using IMU Data

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Abstract

This report presents a variational approach to orientation tracking for a rigid body using Inertial Measurement Unit (IMU) data. We formulate the problem as an optimization task where the objective is to estimate the sequence of unit quaternions that best aligns with sensor measurements. The approach is divided into two stages: sensor calibration and projected gradient descent for trajectory estimation. We propose an alternative *shooting type* unconstrained optimization problem and demonstrate its effectiveness in recovering better orientation estimates compared to direct constrained optimization. Finally, we demonstrate the utility of the estimated orientation by constructing a panoramic image from onboard camera frames.

1 Introduction

State estimation serves as a cornerstone of autonomous robotics. Before an agent can effectively interact with its environment—whether to navigate complex terrains, manipulate objects, or map unknown areas—it must first establish a coherent understanding of its own state, particularly its pose. Since these states are rarely directly observable, they must be inferred from imperfect, noisy sensor measurements. Thus, robust estimation is a critical prerequisite for high-level tasks, enabling precise feedback control, active obstacle avoidance, and safe decision-making under uncertainty.

The problem addressed in this project is the estimation of a rigid body's 3D orientation over time using basic inertial sensors. While the orientation can be integrated directly from angular velocity measurements, sensor bias and noise lead to drift, making simple integration unreliable over long horizons. Our approach formulates this estimation as an optimization problem. Rather than filtering the data sequentially, we seek the trajectory of states (orientations) that best agrees with the sensor measurements globally. The methodology is structured in two parts:

1. **Sensor Calibration:** Estimating intrinsic parameters for the gyroscope and accelerometer using ground truth data.
2. **Pose Optimization:** Minimizing a cost function derived from the motion and observation models to recover the orientation trajectory.

Finally, we apply the estimated orientation trajectory to stitch a series of camera images into a coherent panorama.

Interesting Findings:

- Improved sensor calibration after tuning sensitivity that differs from the datasheet using ground truth data.
- Improved orientation estimates using a *shooting type* unconstrained optimization formulation compared to direct constrained optimization.
- Inherent challenge in accurate yaw estimation due to lack of absolute heading reference from IMU data alone.

2 Problem Formulation

Consider a rigid body equipped with an Inertial Measurement Unit (IMU) and a camera, undergoing pure rotation in 3D space over discrete time steps $t = 0, \ldots, T$. The state of the system is the orientation of the body frame $\{B\}$ relative to the world frame $\{W\}$, represented by a unit quaternion $q_t \in \mathbb{H}_*$, where $\mathbb{H}_* = \{q \in \mathbb{H} \mid \|q\|_2 = 1\}$. Given the angular velocity sequence $\omega_t \in \mathbb{R}^3$ and the time intervals τ_t, the quaternion kinematics are described by

$$q_{t+1} = f(q_t, \tau_t \omega_t) := q_t \circ \exp\left(\left[0, \frac{\tau_t \omega_t}{2}\right]\right), \ t = 0, \ldots, T-1 \tag{1}$$

Furthermore, given the current orientation q_t, the expected accelerometer measurement a_t (assuming no linear acceleration) is given by the gravity vector rotated into the body frame:

$$[0, a_t] = h(q_t) := q_t^{-1} \circ [0, 0, 0, 1] \circ q_t \tag{2}$$

With this notation setup, we define the following problems:

Problem 1 (Sensor Calibration)**.** Let $(q_t)_{t=0}^T$ denote the ground truth orientation trajectory obtained from a VICON motion capture system during a calibration phase. Let $(V_t^{\text{gyro}})_{t=0}^T$ and $(V_t^{\text{acc}})_{t=0}^T$ denote the corresponding raw voltage measurements from the gyroscope and accelerometer, respectively. Let θ^{gyro} and θ^{acc} denote the calibration parameters for the gyroscope and accelerometer, respectively (e.g., biases, sensitivity factors) and let $h^{\text{gyro}}(\cdot)$ and $h^{\text{acc}}(\cdot)$ denote the mapping functions from raw voltages to physical quantities i.e.

$$w_t^{\text{meas}} = h^{\text{gyro}}(V_t^{\text{gyro}}, \theta^{\text{gyro}}) \tag{3}$$
$$a_t^{\text{meas}} = h^{\text{acc}}(V_t^{\text{acc}}, \theta^{\text{acc}}) \tag{4}$$

Then, the Sensor Calibration problem is to find the calibration parameters $\hat{\theta} = (\hat{\theta}^{\text{gyro}}, \hat{\theta}^{\text{acc}})$ that minimize the discrepancy between the measured quantities and those predicted by the ground truth orientations.

Problem 2 (Orientation Tracking)**.** Given the calibrated physical measurements ω_t^{meas} and a_t^{meas} (derived from $\hat{\theta}$) and an initial orientation q_0, estimate the trajectory of orientations $\hat{q}_{1:T}$ that satisfies the quaternion kinematics (1) and agrees with the IMU measurments.

Problem 3 (Panorama Construction)**.** Given the estimated orientation trajectory $\hat{q}_{1:T}$ and a sequence of RGB images $\{I_t\}_{t=0}^T$ captured by the onboard camera, construct a single panoramic image P by mapping all pixels (u, v) from individual image frames I_t onto a global cylindrical coordinate system (y, θ) utilizing the rotation determined by $\hat{q}_t$.

3 Technical Approach

Our solution follows a sequential pipeline: first, we map raw sensor voltages to physical units using a calibrated model; second, we optimize the full orientation trajectory to minimize motion and observation errors; finally, we use the estimated orientations to stitch images into a panorama.

3.1 Sensor Calibration

The IMU outputs raw voltages which are related to the physical quantities of interest via scale factors and biases. The mapping functions h^{gyro} and h^{acc} in (3) and (4)are defined as:

$$\omega_t^{\text{meas}}(\theta^{\text{gyro}}) = h^{\text{gyro}}(V_t^{\text{gyro}}, \theta^{\text{gyro}}) = \frac{V_t^{\text{gyro}} - V_{\text{ref}}^{\text{gyro}}}{k^{\text{gyro}}} - b^{\text{gyro}} \tag{5}$$

$$a_t^{\text{meas}}(\theta^{\text{acc}}) = h^{\text{acc}}(V_t^{\text{acc}}, \theta^{\text{acc}}) = \frac{V_t^{\text{acc}} - V_{\text{ref}}^{\text{acc}}}{k^{\text{acc}}} - b^{\text{acc}} \tag{6}$$

where the parameter sets are defined as $\theta^{\text{gyro}} = \{V_{\text{ref}}^{\text{gyro}}, k^{\text{gyro}}, b^{\text{gyro}}\}$ and $\theta^{\text{acc}} = \{V_{\text{ref}}^{\text{acc}}, k^{\text{acc}}, b^{\text{acc}}\}$. According to the datasheet specifications, we set the sensitivity parameters $k^{\text{gyro}} = 3.33$ mV/deg/s (operating in 4X amplification mode) and $k^{\text{acc}} = 300$ mV/g and the reference voltages $V_{\text{ref}}^{\text{gyro}} = 1.23$ V and $V_{\text{ref}}^{\text{acc}} = 1.5$ V. However, these values may deviate in practice due to manufacturing tolerances and environmental factors, necessitating calibration.

Estimating the Bias: We estimate the biases b^{gyro} and b^{acc} using the static portions of the calibration dataset where the IMU is stationary. We have knowledge that the first $T_{\text{stat}} = 700$ time steps correspond to a static state with $q_t = [1, 0, 0, 0]$ (no rotation). Thus, we compute the biases as:

$$b^{\text{gyro}} = \frac{1}{T_{\text{stat}}} \sum_{t=0}^{T_{\text{stat}}-1} \left(\frac{V_t^{\text{gyro}} - V_{\text{ref}}^{\text{gyro}}}{k^{\text{gyro}}} \right) \tag{7}$$

$$b^{\text{acc}} = \frac{1}{T_{\text{stat}}} \sum_{t=0}^{T_{\text{stat}}-1} \left(\frac{V_t^{\text{acc}} - V_{\text{ref}}^{\text{acc}}}{k^{\text{acc}}} - [0, 0, 1] \right) \tag{8}$$

Verifying Calibration: We can compute the predicted linear acceleration from the ground truth orientations and compare it to the calibrated accelerometer measurements. The predicted acceleration in the body frame is given by equation (2). Hence, we expect:

$$[0, a_t^{\text{meas}}(\theta^{\text{acc}})] \approx h(q_t) = q_t^{-1} \circ [0, 0, 0, 1] \circ q_t, \quad t = 1, \ldots, T \tag{9}$$

We cannot directly verify the gyroscope calibration since we do not have ground truth angular velocity measurements. However, we can integrate the calibrated angular velocities to obtain an orientation trajectory and compare it to the ground truth orientations. Let $q_{1:T}^{\text{meas}}$ denote the trajectory obtained by integrating the calibrated angular velocities starting from q_0. We expect:

$$q_{t+1} \approx q_{t+1}^{\text{meas}} = f(q_t^{\text{meas}}, \tau_t \omega_t^{\text{meas}}(\theta^{\text{gyro}})), \quad t = 0, \ldots, T-1 \tag{10}$$

Based on these verifications, we can iteratively adjust the sensitivity parameters k^{gyro} and k^{acc} to minimize the discrepancies observed in the above comparisons, thereby refining our calibration. Note that the biases are re-estimated after each sensitivity adjustment and that tuning the reference voltages $V_{\text{ref}}^{\text{gyro}}$ and $V_{\text{ref}}^{\text{acc}}$ is not necessary, because the bias terms effectively absorb any constant offset. This calibration procedure is performed manually by iteratively adjusting the sensitivity parameters and re-evaluating the discrepancies until satisfactory alignment with ground truth is achieved.

Results: We observe significant improvements in alignment with ground truth data after tuning the sensitivity parameters away from their nominal datasheet values. For instance, in dataset 2, adjusting the accelerometer sensitivity from 300 mV/g to 100 mV/g and the gyroscope sensitivity from 3.33 mV/deg/s to 0.83 mV/deg/s resulted in markedly better agreement with ground truth accelerations and orientations, respectively. Figures 1 and 2 illustrate these improvements, showcasing the calibrated measurements closely following the ground truth data post-calibration. These adjustments work well accross all datasets provided, the final sensitivity values used are

$$k^{\text{acc}} = 100 \text{ mV/g}, \quad k^{\text{gyro}} = 0.83 \text{ mV/deg/s}. \tag{11}$$

3.2 Orientation Tracking

With the calibrated measurements ω_t^{meas} and a_t^{meas} fixed, we seek the estimated orientation trajectory $\hat{q}_{1:T}$ that combines the sensor measurments and the motion model.

3.2.1 Approach 1: Constrained Optimization

We formulate the state estimation problem as an optimization problem minimizing a cost function that penalizes deviations from both the motion and observation models. The process cost penalizes the difference between the current state $\hat{q}_{t+1}$ and the state predicted by applying the measured angular velocity to $\hat{q}_t$. Using the logarithmic map to measure distances in the tangent space of the quaternion manifold, this term is:

$$J_{\text{motion}} = \frac{1}{2} \sum_{t=0}^{T-1} \left\| 2 \log \left(\hat{q}_{t+1}^{-1} \circ f(\hat{q}_t, \tau_t \omega_t^{\text{meas}}) \right) \right\|_2^2 \tag{12}$$

The measurement cost penalizes the difference between the measured acceleration and the expected gravity direction in the body frame:

$$J_{\text{obs}} = \frac{1}{2} \sum_{t=1}^{T} \| [0, a_t^{\text{meas}}] - h(\hat{q}_t) \|_2^2 \tag{13}$$

We solve the constrained optimization problem:

$$\begin{aligned} \min_{\hat{q}_{1:T}} \quad & J_{\text{motion}}(\hat{q}_{1:T}) + J_{\text{obs}}(\hat{q}_{1:T}) \\ \text{s.t.} \quad & \|\hat{q}_t\|_2 = 1, \quad t = 1, \dots, T \\ & \hat{q}_0 = q_0 \end{aligned} \tag{14}$$

We employ a Projected Gradient Descent (PGD) scheme. At each iteration k, we update the trajectory using the gradient of the cost function ∇J and a step size α:

$$\hat{q}_t^{(k+1)} \leftarrow \hat{q}_t^{(k)} - \alpha \nabla_{\hat{q}_t} J(\hat{q}_{1:T}^{(k)}) \tag{15}$$

After each update, the constraint is enforced by normalizing the quaternions:

$$\hat{q}_t^{(k+1)} \leftarrow \frac{\hat{q}_t^{(k+1)}}{\|\hat{q}_t^{(k+1)}\|_2} \tag{16}$$

The gradient $\nabla_{\hat{q}_t} J$ is computed efficiently using standard backpropagation using pytorch's autograd functionality.

Warm Starting: We initialize the optimization using q^{meas} from equation (10). This is the trajectory obtained by integrating the calibrated angular velocities from the initial orientation q_0. This provides a reasonable starting point that respects the motion model, allowing the optimization to focus on correcting drift and aligning with the accelerometer measurements.

3.2.2 Approach 2: Shooting type Optimization

An alternative approach is to formulate the optimization problem in a *shooting type* manner. Instead of directly optimizing over the entire trajectory $\hat{q}_{1:T}$, we optimize over the angular velocity vectors. Let $(\hat{\omega}_t)_{t=0}^{T-1}$ denote the sequence of angular velocities to be optimized. The orientation trajectory is then generated by integrating these angular velocities starting from the initial orientation q_0:

$$\hat{q}_{t+1} = f(\hat{q}_t, \tau_t \hat{\omega}_t), \quad t = 0, \dots, T-1 \tag{17}$$

The cost function now depends on $(\hat{\omega}_t)_{t=0}^{T-1}$ through the generated trajectory $\hat{q}_{1:T}$. We propose the following unconstrained optimization problem:

$$\begin{aligned} \min_{\hat{\omega}_{0:T-1}} \quad & \frac{1}{2}\sum_{t=0}^{T-1} \|\hat{w}_t - w_t^{\text{meas}}\|_2^2 + \frac{1}{2}\sum_{t=1}^{T} \|[0, a_t^{\text{meas}}] - h(\hat{q}_t)\|_2^2 \\ \text{s.t.} \quad & \hat{q}_{t+1} = f(\hat{q}_t, \tau_t \hat{\omega}_t), \quad t = 0, \dots, T-1 \\ & \hat{q}_0 = q_0 \end{aligned} \tag{18}$$

This is inspired from the optimal control literature where the control inputs (angular velocities) are optimized to minimize a cost function over the resulting state trajectory. The name *shooting* comes from the idea of *shooting* forward from the initial state using the control inputs to generate the trajectory. The cost function includes a term penalizing deviations of the optimized angular velocities from the measured ones, in addition to the accelerometer observation cost. This formulation avoids the need for explicit constraints on the quaternions, as the unit norm is inherently preserved through the quaternion integration process.

Advantages and Drawbacks: The shooting type formulation is unconstrained, which allows it to find better minima since it is not restricted by the projection step. However, it is computationally more intensive since each evaluation of the cost function requires integrating the entire trajectory from the angular velocities. In contrast, the constrained optimization directly optimizes the trajectory, making it more efficient per iteration but potentially less effective at finding the global minimum due to the projection step. Furthermore, the shooting type approach allows for direct regularization of the angular velocities, which can be beneficial in reducing noise-induced fluctuations.

Dataset	Vanilla RMS	Modified RMS
1	1.7102	**0.9913**
2	1.3983	**0.5166**
3	1.9651	**1.7849**
4	0.8957	**0.6467**
5	2.7250	**2.7206**
8	1.2551	**1.1825**
9	1.0776	**0.8222**

Table 1: Comparison of RMS errors in Euler angles. The shooting type optimization (Modified RMS) consistently outperforms the constrained approach (Vanilla RMS) across all datasets.

A measure of performance: We evaluate the orientation tracking performance by computing the Root Mean Square (RMS) error between the estimated Euler angles (roll, pitch, yaw) derived from the estimated quaternions and the ground truth Euler angles obtained from the VICON system. This metric provides a quantitative measure of how closely the estimated orientations align with the true orientations over the entire trajectory. We summarize the RMS errors for both optimization approaches across all datasets in a comparative table in Table 1.

Results: The shooting type optimization consistently yields lower RMS errors across all datasets compared to the constrained optimization approach. Figures 3(a) and 3(b) illustrate the orientation tracking results for datasets 2 and 9, respectively.

3.3 Panorama Construction

The final stage of our pipeline utilizes the estimated orientation trajectory $\hat{q}_{1:T}$ to fuse the sequence of onboard camera images into a single panoramic mosaic. This process involves projecting the local camera pixels into a global cylindrical coordinate system. The procedure is executed in the following steps:

1. Ray Generation (Spherical to Cartesian): For each pixel (u, v) in a camera frame I_t, we first compute its corresponding viewing ray in the camera frame. Assuming a pinhole camera model with horizontal and vertical fields of view (FOV) θ_h and θ_v, we map the pixel coordinates to spherical angles (λ_c, ϕ_c):

$$\lambda_c = \left(\frac{u}{W} - 0.5\right)\theta_h, \quad \phi_c = -\left(\frac{v}{H} - 0.5\right)\theta_v \tag{19}$$

These angles are then converted into a 3D unit vector $\mathbf{x}_c \in \mathbb{R}^3$ in the camera's Cartesian coordinate system:

$$\mathbf{x}_c = \begin{bmatrix} \cos\phi_c \sin\lambda_c \\ \sin\phi_c \\ \cos\phi_c \cos\lambda_c \end{bmatrix} \tag{20}$$

where y corresponds to the vertical axis in our convention.

2. Transformation (Camera(t) to Camera(0)): The camera-frame vector at time t is transformed into the camera frame vector at time 0 using the estimated body orientation $\hat{q}_t$ and the camera position $p = [0, 0, 0.1]$ with respect to the body frame. Since the camera is rigidly attached to the body, we account for the extrinsic calibration R_{cam2body}. This means that $T_{B_t,C_t} = (R_{\text{cam2body}}, p)$ maps camera frame vectors at time t to the body frame at time t. We are interested in mapping camera frame vectors at time t to the camera frame at time $t = 0$, which is given by

$$\begin{aligned} T_{C_0,C_t} &= (T_{B_0,C_0})^{-1}\, T_{B_0,B_t}\, T_{B_t,C_t} \\ &= \begin{bmatrix} R_{\text{cam2body}}^T & -R_{\text{cam2body}}^T p \\ 0 & 1 \end{bmatrix} \begin{bmatrix} R & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} R_{\text{cam2body}} & p \\ 0 & 1 \end{bmatrix} \\ &= \begin{bmatrix} R_{\text{cam2body}}^T R\, R_{\text{cam2body}} & R_{\text{cam2body}}^T (Rp - p) \\ 0 & 1 \end{bmatrix} \end{aligned}$$

where R is the rotation matrix corresponding to the quaternion $\hat{q}_t$. Applying this transformation to the camera frame vector $\mathbf{x}_c$ gives us the world-frame (camera frame at $t = 0$) vector $\mathbf{x}_w$:

$$[\mathbf{x}_w, 1] = T_{C_0,C_t}[\mathbf{x}_c, 1] \tag{21}$$

3. Projection (Cartesian to Spherical to Cylindrical): To map the 3D world vectors onto a 2D panoramic canvas, we convert $\mathbf{x}_w = [x_w, y_w, z_w]^T$ back to global spherical coordinates (λ_w, ϕ_w):

$$\lambda_w = \text{atan2}(x_w, z_w), \quad \phi_w = \arcsin\left(\frac{y_w}{\sqrt{x_w^2 + y_w^2 + z_w^2}}\right) \tag{22}$$

Finally, we apply a cylindrical projection where the horizontal coordinate on the panorama u_{pan} and the vertical coordinate v_{pan} are given by (assuming depth = 1)

$$u_{\text{pan}} = \lambda_w, \quad v_{\text{pan}} = \phi_w \tag{23}$$

This mapping allows us to place the RGB values from the source image I_t onto the global panorama canvas $P(u_{\text{pan}}, v_{\text{pan}})$.

Results: The quality of the constructed panorama is highly dependent on the accuracy of the estimated orientation trajectory. In dataset 8 and 9, we observe that the panorama generated using orientations from the shooting type optimization exhibits better alignment and less distortion compared to the panorama generated from the constrained optimization. The shooting type approach captures the overall orientation trajectory more accurately, resulting in a panoramic image that closely resembles the one constructed using VICON ground truth orientations, as shown in Figure 5 and Figure 6. Finally, we generate panoramas for all test datasets using the shooting type optimization results, which demonstrate coherent stitching and minimal distortion, further validating the effectiveness of our orientation estimation approach. The results are demonstrated in Figures 7 and 8. We must note that in 8, the panorama is pretty bad when the yaw changes significantly, which is consistent with the fact that yaw estimation is the most challenging aspect of the orientation tracking problem due to the lack of an absolute heading reference from the IMU data alone. Furthermore, the spherical projection used in our pipeline may not be ideal for large yaw changes, as it can lead to significant distortion in the projected image.

4 Results

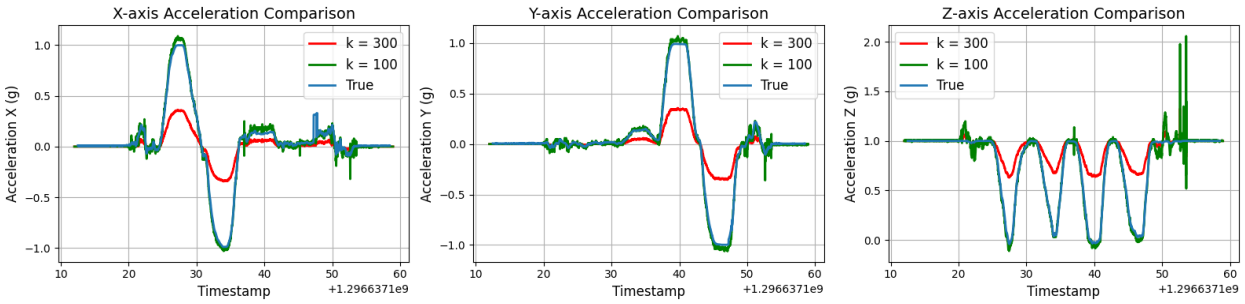
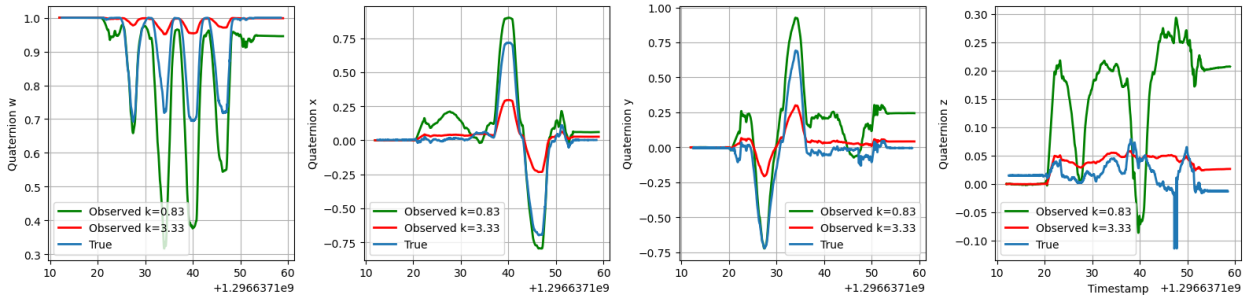
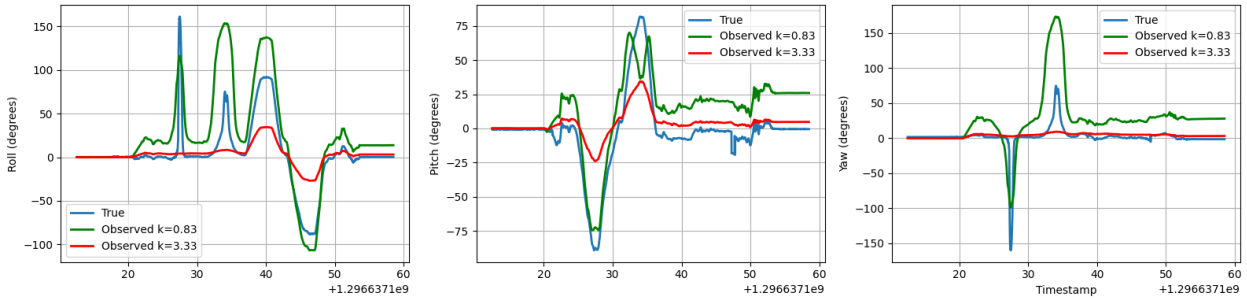


Figure 1: Accelerometer calibration for dataset 2. The sensitivity k^{acc} was adjusted from the nominal value of 300 mV/g (red) to 100 mV/g (green) to improve alignment with ground truth accelerations (blue).

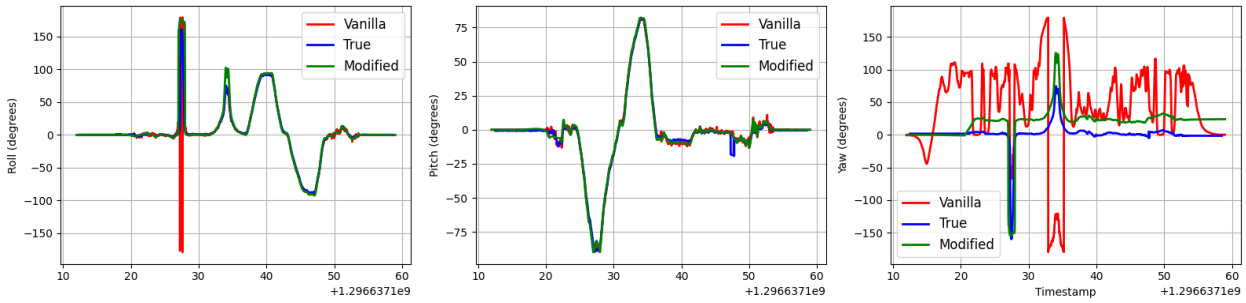


(a) Quaternion comparison after gyroscope calibration

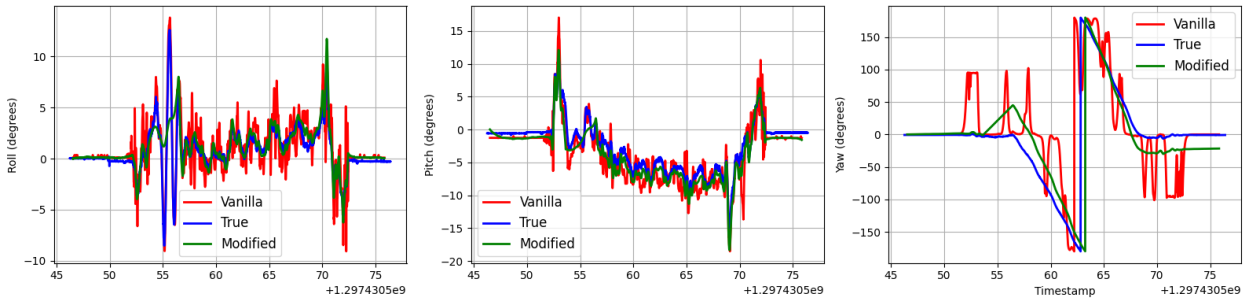


(b) Euler angle comparison after gyroscope calibration

Figure 2: Gyroscope calibration for dataset 2. The sensitivity k^{gyro} was adjusted from the nominal value of 3.33 mV/deg/s (red) to 0.83 mV/deg/s (green) to improve alignment with ground truth orientations (blue).

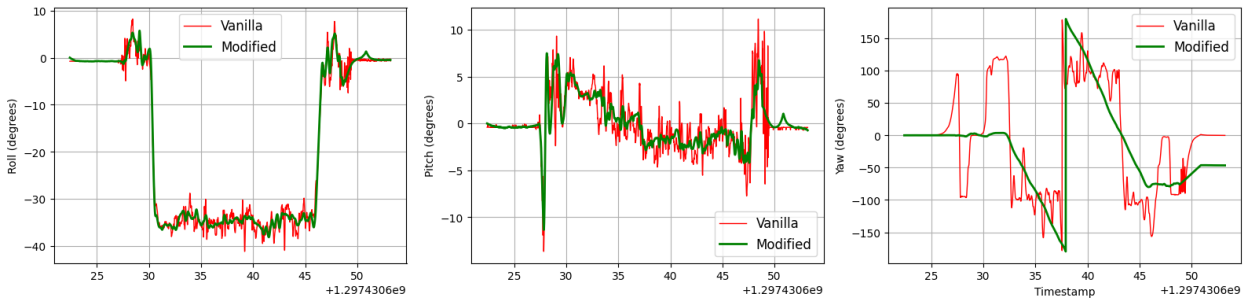


(a) Dataset 2: Orientation tracking

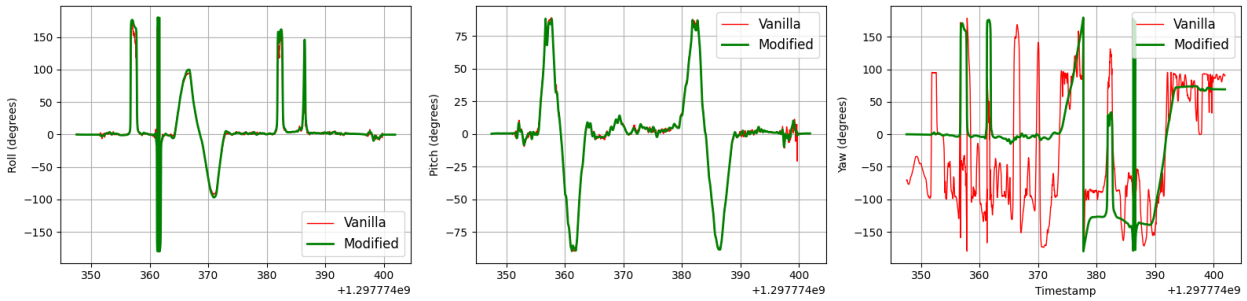


(b) Dataset 9: Orientation tracking

Figure 3: Orientation tracking results comparing the proposed shooting type optimization (green) against the constrained optimization (red) and ground truth (blue) for datasets 2 and 9. The shooting type approach consistently yields lower RMS errors across all datasets. In (a), the constrained optimization struggles with yaw estimation, while the shooting type method better captures the overall orientation trajectory. In (b), we can see improved noise reduction in roll and pitch estimates using the shooting type approach.



(a) Dataset 10: Orientation tracking

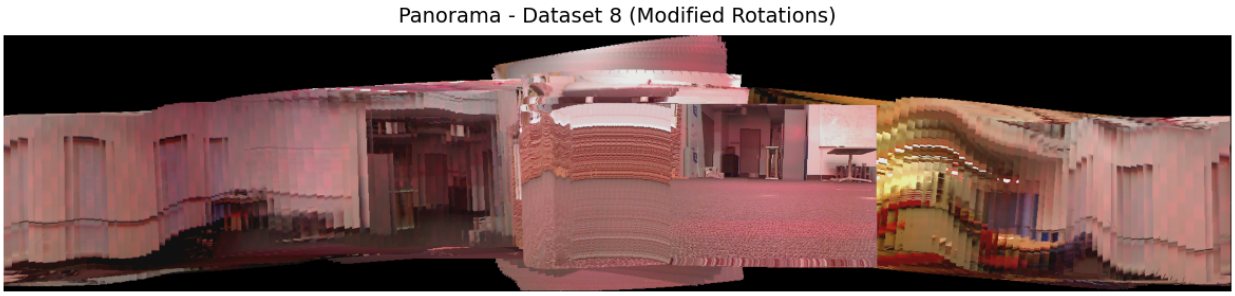


(b) Dataset 11: Orientation tracking

Figure 4: Orientation tracking results comparing the proposed shooting type optimization (green) against the constrained optimization (red) for test datasets 10 and 11. In (a), we can see that the modified approach has less variance than the vanilla approach. In (b), the modified approach seemse to better capture the yaw changes in the trajectory.



(a) Dataset 8: Panorama using constrained optimization



(b) Dataset 8: Panorama using shooting optimization



(c) Dataset 8: Panorama using VICON ground truth

Figure 5: Panoramic image construction for dataset 8 using orientations estimated from the constrained optimization (a), the shooting type optimization (b), and the VICON ground truth (c). The panorama generated using the shooting type optimization exhibits better alignment and less distortion compared to the constrained optimization, closely resembling the panorama constructed from the ground truth orientations.



(a) Dataset 9: Panorama using constrained optimization

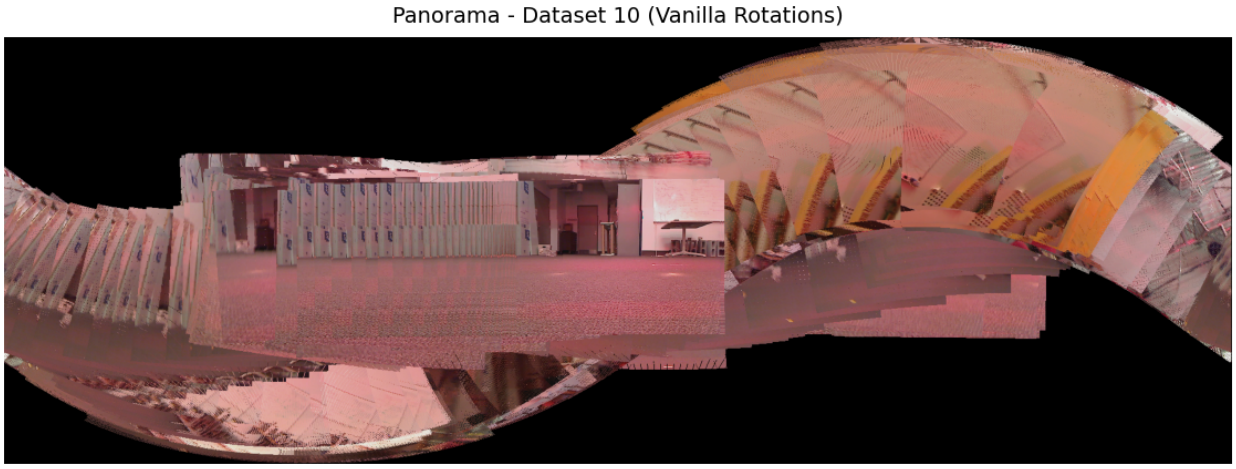


(b) Dataset 9: Panorama using shooting optimization

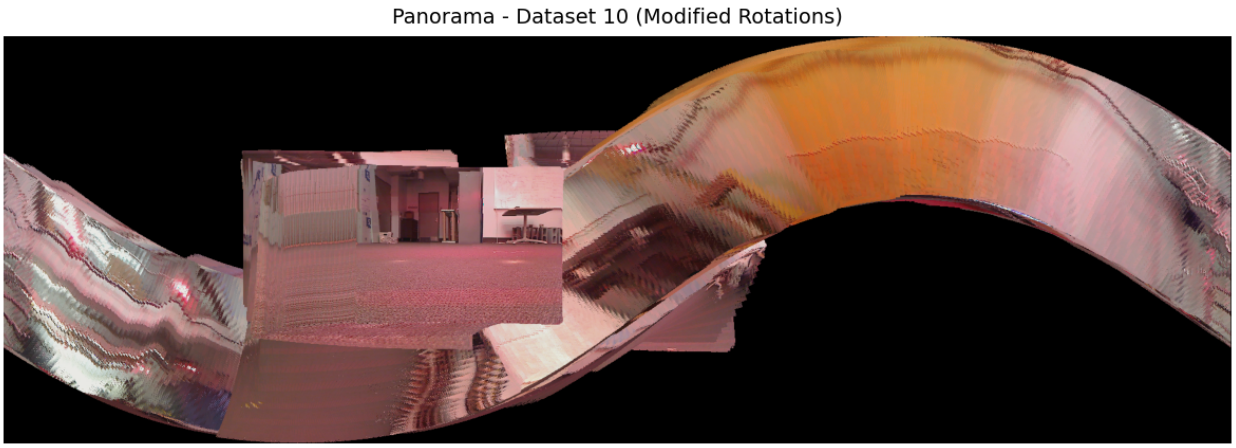


(c) Dataset 9: Panorama using VICON ground truth

Figure 6: Panoramic image construction for dataset 9 using orientations estimated from the constrained optimization (a), the shooting type optimization (b), and the VICON ground truth (c). The panorama generated using the shooting type optimization exhibits better alignment and less distortion compared to the constrained optimization, closely resembling the panorama constructed from the ground truth orientations.

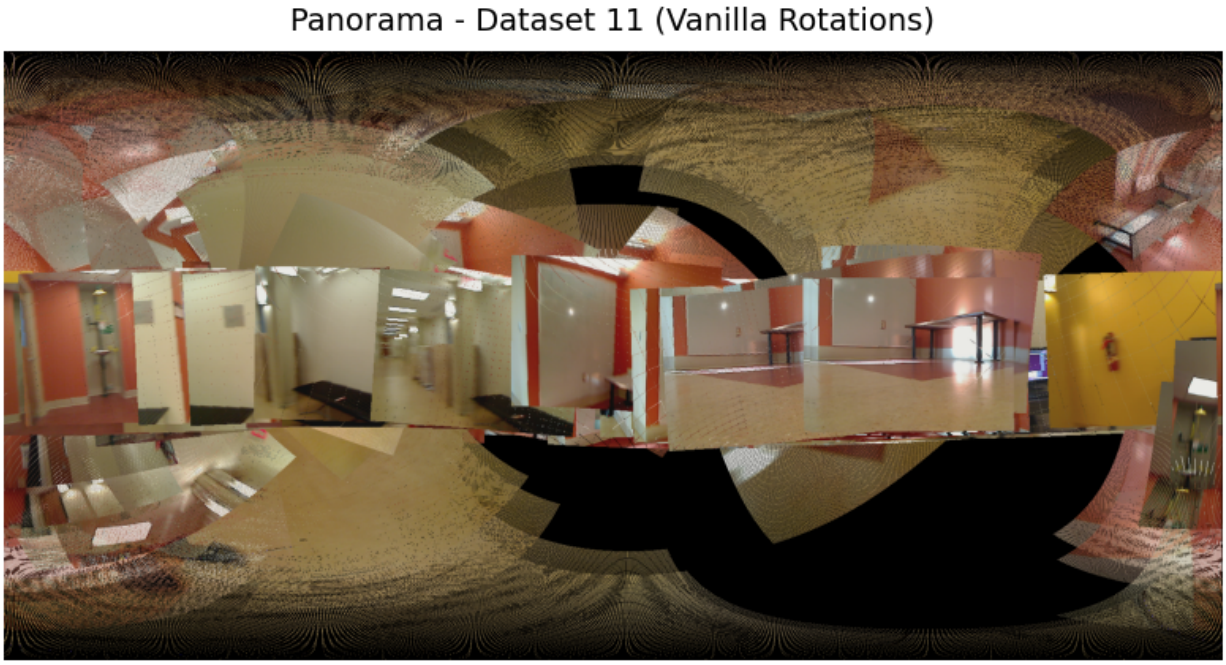


(a) Dataset 10: Panorama using constrained optimization

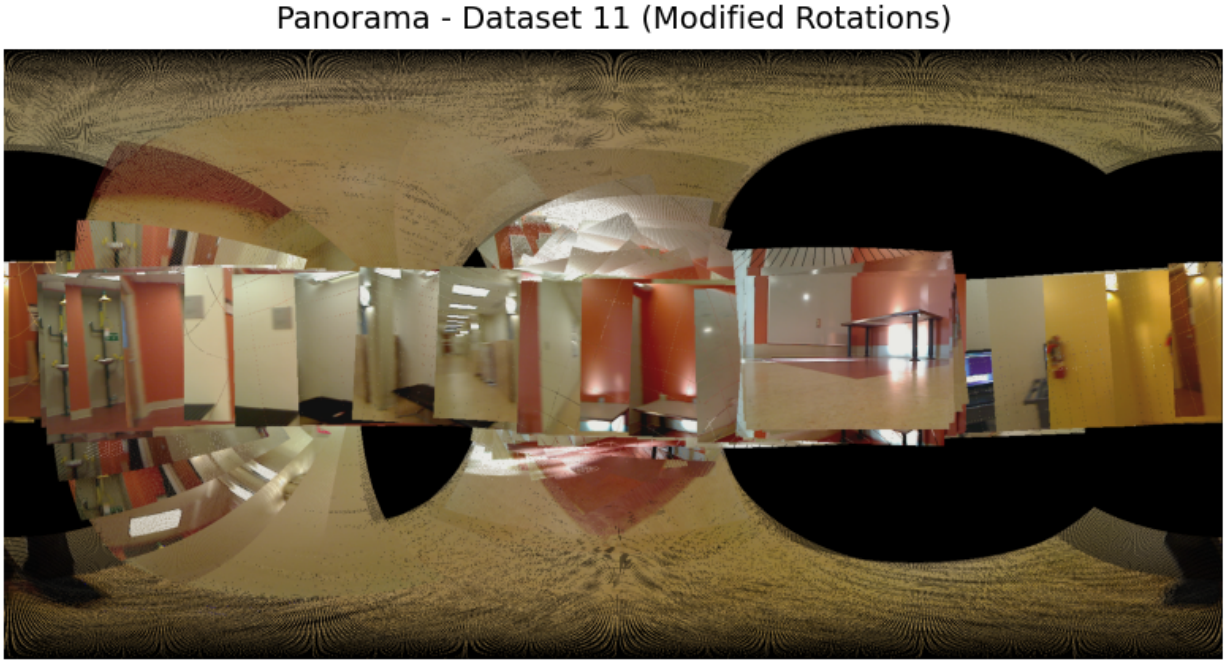


(b) Dataset 10: Panorama using shooting optimization

Figure 7: Panoramic image construction for dataset 10 using orientations estimated from the constrained optimization (a) and the shooting type optimization (b). The panorama generated using the shooting type optimization exhibits better continuity and less distortion compared to the constrained optimization.



(a) Dataset 11: Panorama using constrained optimization



(b) Dataset 11: Panorama using shooting optimization

Figure 8: Panoramic image construction for dataset 11 using orientations estimated from the constrained optimization (a) and the shooting type optimization (b). The panorama generated using the shooting type optimization exhibits better continuity and less distortion compared to the constrained optimization. However, the overall quality of the panorama is lower than in dataset 10, likely due to the more complex yaw motion and increased noise in the orientation estimates for this dataset.